

SESSION 3

Regression Fundamentals

Linear models • Polynomial terms • Regularization • Residuals • Regression metrics

Applied AI Design Workshop for Faculty

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Goal: build intuition for predicting continuous outcomes and judging model error.

Session focus

- Regression problem framing
- Baselines and model fit
- Linear vs. nonlinear terms
- Regularization
- Residuals and metrics

Session 3

From engineering problem to ML problem

Classic ML overview

Invited presentation: From Prompts to Agents

Break

Regression fundamentals

Classification fundamentals

Hands-on baseline model activity

Earlier
Problem framing
Task choice

Now

Next

Practice

Session 3 learning outcomes

Define

Recognize when a continuous target makes regression appropriate

Model

Explain linear, polynomial, and regularized regression at a conceptual level

Evaluate

Compare model performance to a simple baseline using common metrics

Diagnose

Use residuals to identify bias, outliers, nonlinearity, or heteroscedasticity

Discuss

Decide when more complex models are justified for disciplinary use

Participant output: a regression problem statement, a baseline choice, an initial metric, and one diagnostic question about model error.

What makes a problem a regression problem?

Regression predicts a continuous numerical outcome. The model is judged by how far predictions are from observed values.

Use regression when asking:

- How much?
- How many?
- What value?
- What final measurement?
- What future quantity?

Typical targets:

- Strength
- Energy use
- Cycle time
- Yield
- Temperature
- Score
- Remaining life

Not regression when:

- Target is a class label
- Goal is discovering groups
- Output is open-ended text
- The main goal is causal proof

Key discipline question: What error magnitude is acceptable for this application?

Activity: write a regression problem statement

Use the template to convert a disciplinary problem into a regression task.

Using _____ as inputs, develop a regression model to predict _____ for each _____, with performance evaluated using _____.

Inputs

Variables available
before prediction

Target

Continuous outcome
to predict

Unit

Run, sample, student,
machine cycle,
patient, day

Metric

MAE, RMSE, R^2 , or
domain threshold

Quick check: are all input variables available at the time the prediction must be made?

Example: predicting material strength

A regression dataset pairs process and material conditions with a continuous measured outcome.

Run	Temp	Pressure	Time	Cooling	Strength
1	185	42	28	Slow	72.4
2	190	45	30	Fast	75.8
3	175	39	24	Slow	65.1
4	205	47	35	Fast	82.6
5	198	44	31	Slow	79.3

Prediction objective
Estimate final strength before destructive testing.

Baseline
Predict the mean training-set strength for every run.

Metric
Use MAE when errors are easy to interpret in original units.

Start with a baseline before training a model

A baseline is the simplest prediction that a machine learning model must improve upon.

Mean baseline

Predict the average training target for every new case.

Useful when squared error or RMSE is central.

Median baseline

Predict the median training target for every new case.

Useful when outliers may distort the mean.

Domain baseline

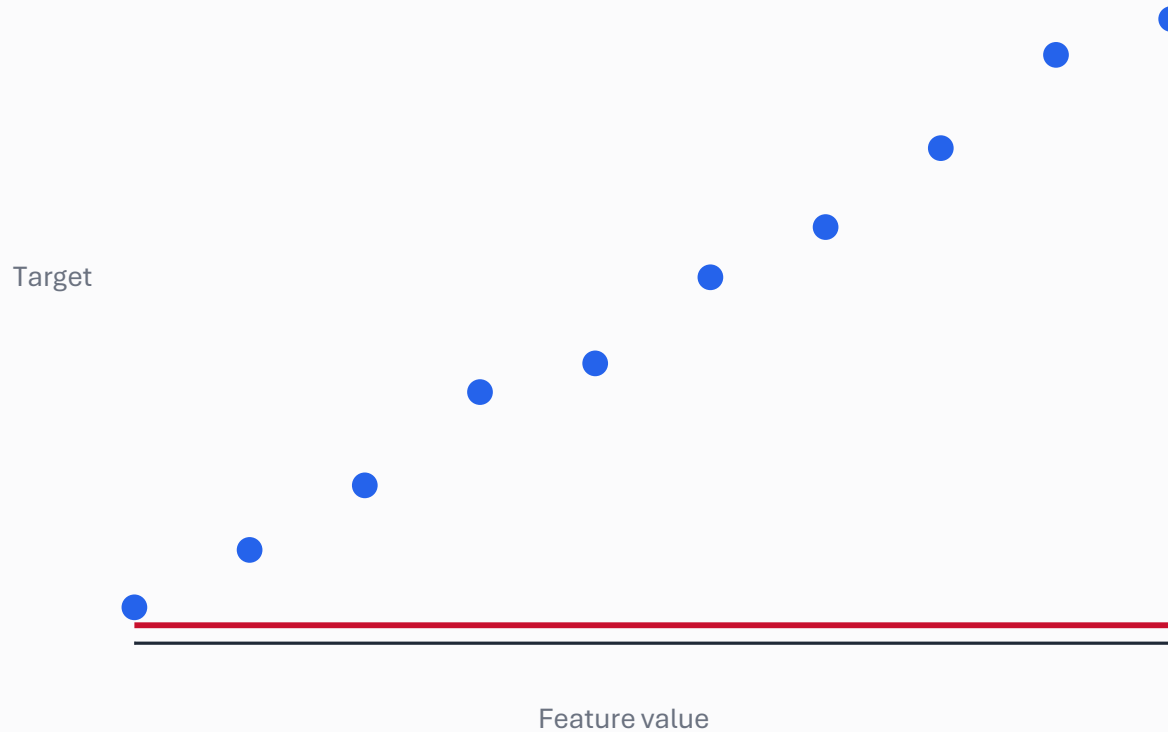
Compare to current practice, heuristic, lookup table, or physics estimate.

Often the most persuasive baseline.

Discussion prompt: Is “better than the mean” enough, or must the model beat current disciplinary practice?

Linear regression: a first interpretable model

Linear regression estimates how the target changes with input features, assuming an approximately linear relationship.



Model form

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

Interpretation

A coefficient estimates average target change when one feature changes, holding others constant.

Caution

Interpretation depends on data quality, feature scaling, correlation, and study design.

Linear regression: a first interpretable model

In ML, we write the equation for a linear regression model as follows:

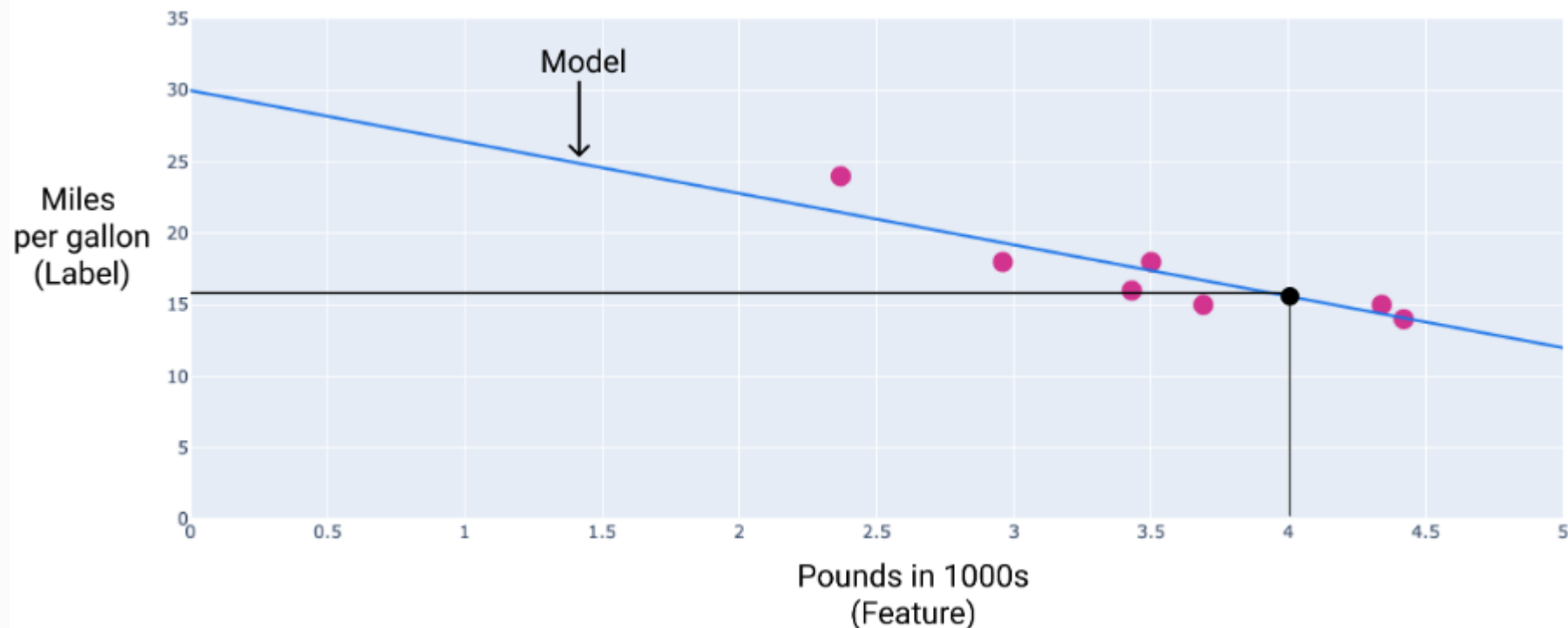
$$y' = b + w_1 x_1$$

where:

- **y'** is the predicted label—the output.
- **b** is the **bias** of the model. Bias is the same concept as the y-intercept in the algebraic equation for a line. In ML, bias is sometimes referred to as w_0 . Bias is a **parameter** of the model and is calculated during training.
- **w₁** is the **weight** of the feature. Weight is the same concept as the slope m in the algebraic equation for a line. Weight is a **parameter** of the model and is calculated during training.
- **x₁** is a **feature**—the input.

Linear regression: Training Model

In our example, we'd calculate the weight and bias from the line we drew. The bias is 30 (where the line intersects the y-axis), and the weight is -3.6 (the slope of the line). The model would be defined as $y' = 30 + (-3.6)(x_1)$, and we could use it to make predictions. For instance, using this model, a 4,000-pound car would have a predicted fuel efficiency of 15.6 miles per gallon.

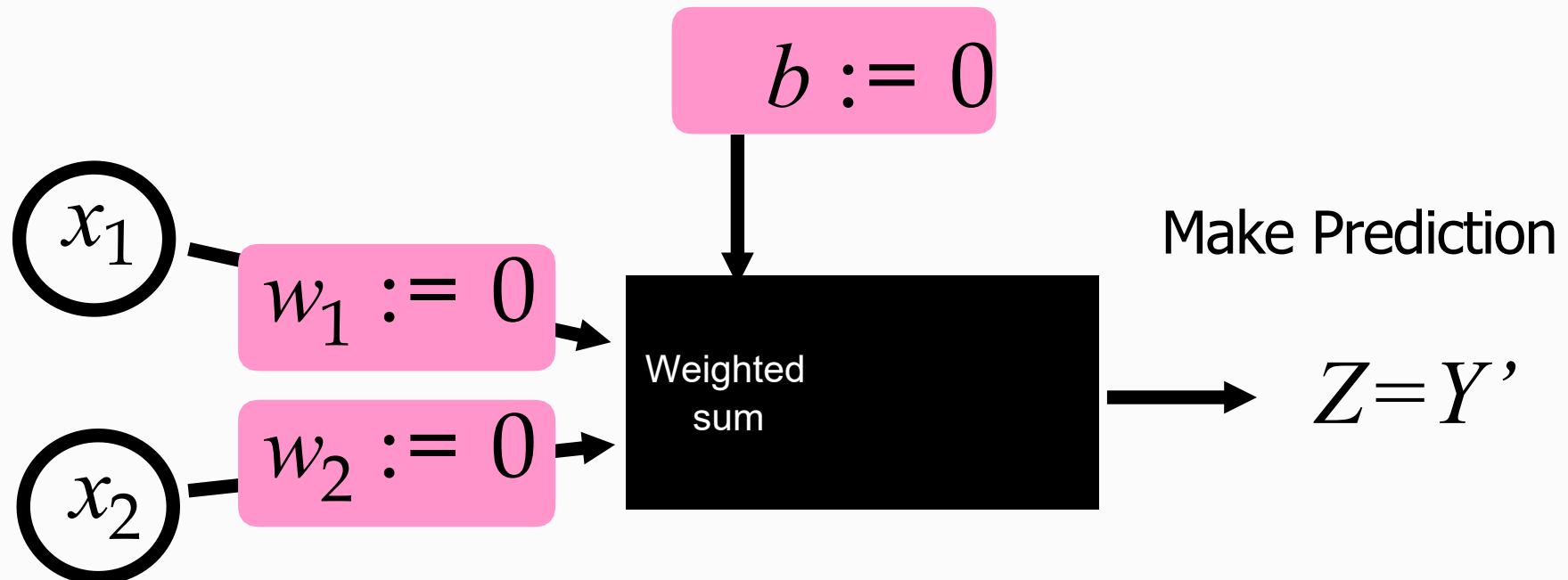


Using the model, a 4,000-pound car has a predicted fuel efficiency of 15.6 miles per gallon.

Linear regression: a first interpretable model

1. Define training set
2. Initialize model weights and bias to zero

Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
3.44	18
3.43	16
4.34	15
4.42	14
2.37	24



Linear regression: a first interpretable model

Model parameters to compute value

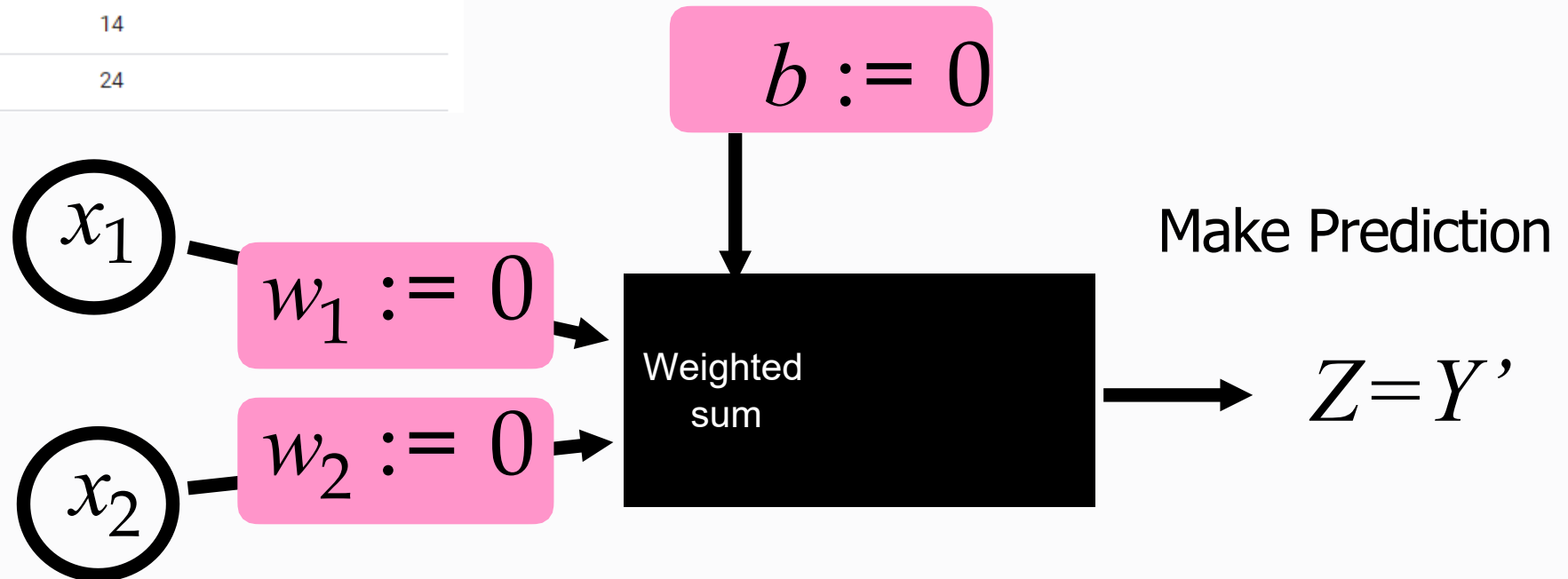
$$\bigcirc x_1 \times \textcolor{pink}{w_1} + \bigcirc x_2 \times \textcolor{pink}{w_2} + \textcolor{pink}{b} = z = y'$$

$$x_1 \times w_1 + x_2 \times w_2 + \dots + x_m \times w_m + b = z = y'$$

Linear regression: a first interpretable model

Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
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What is the model first prediction

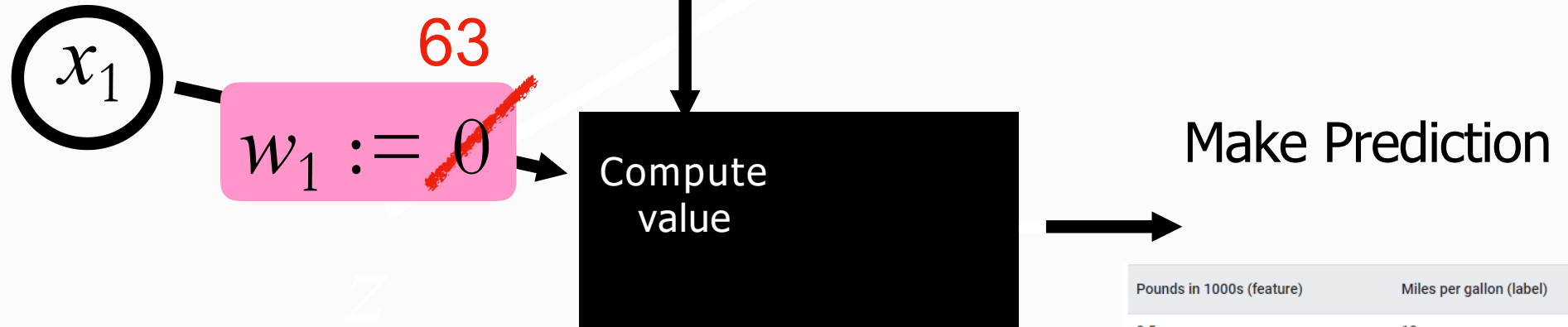


Linear regression: a first interpretable model

Update the weights based on the error

For where $x_1 = 3.5$, and $b = 0$, and Error is 18

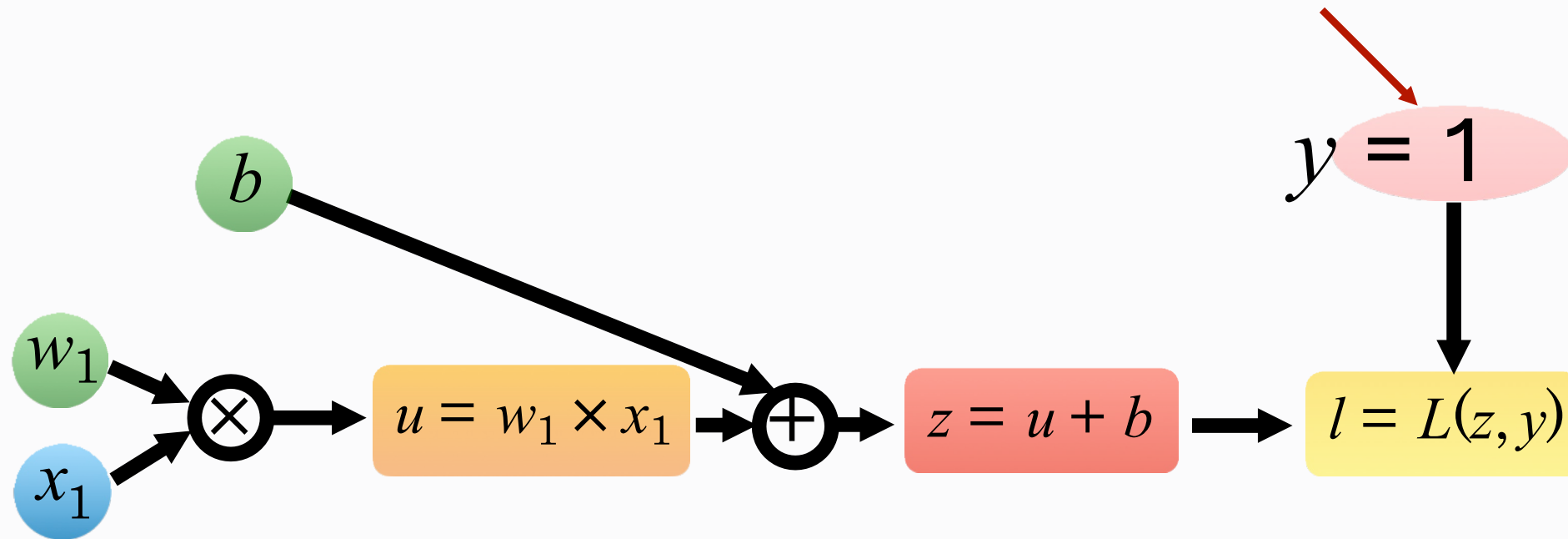
$$w_j := w_j + error \times x_j^{[i]}$$
$$b := b + error$$



Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
3.44	18
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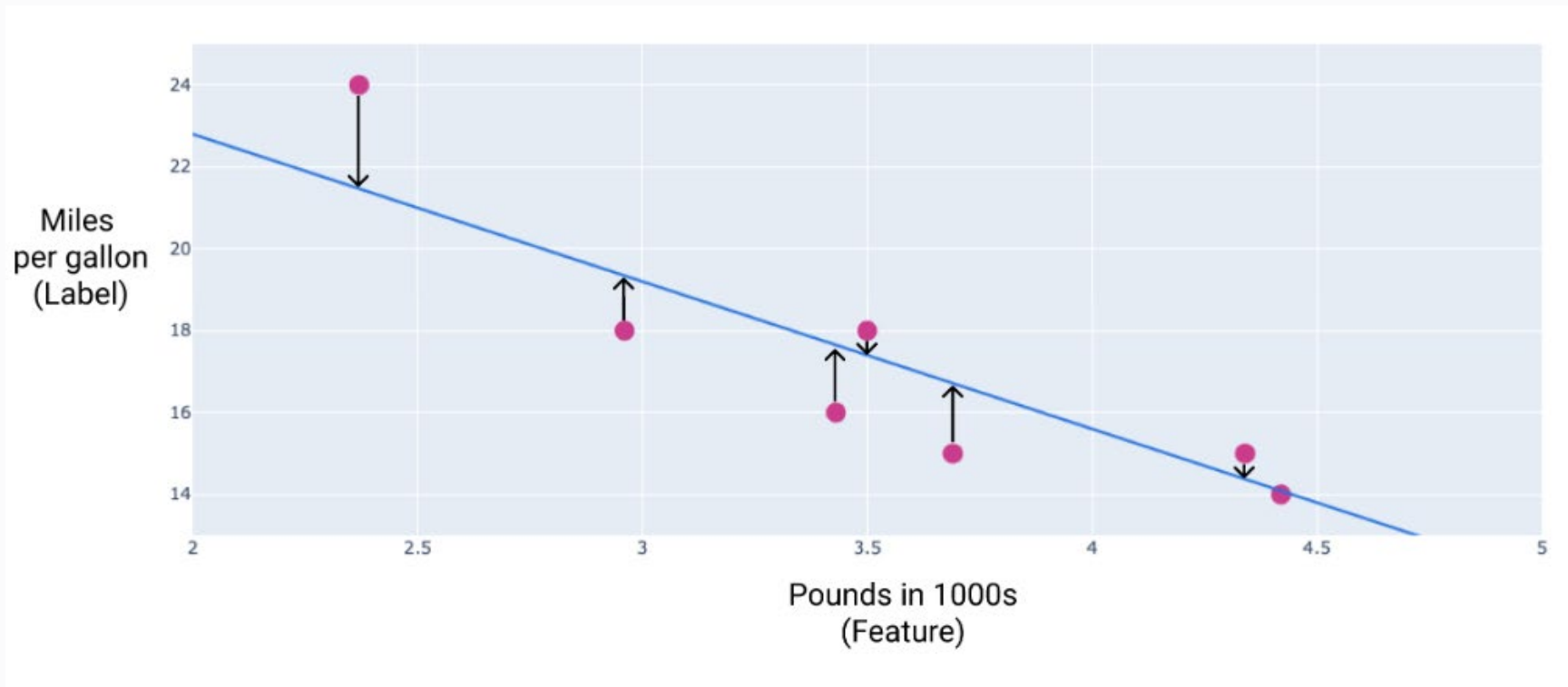
Linear regression: a first interpretable model

Computation Graph of System



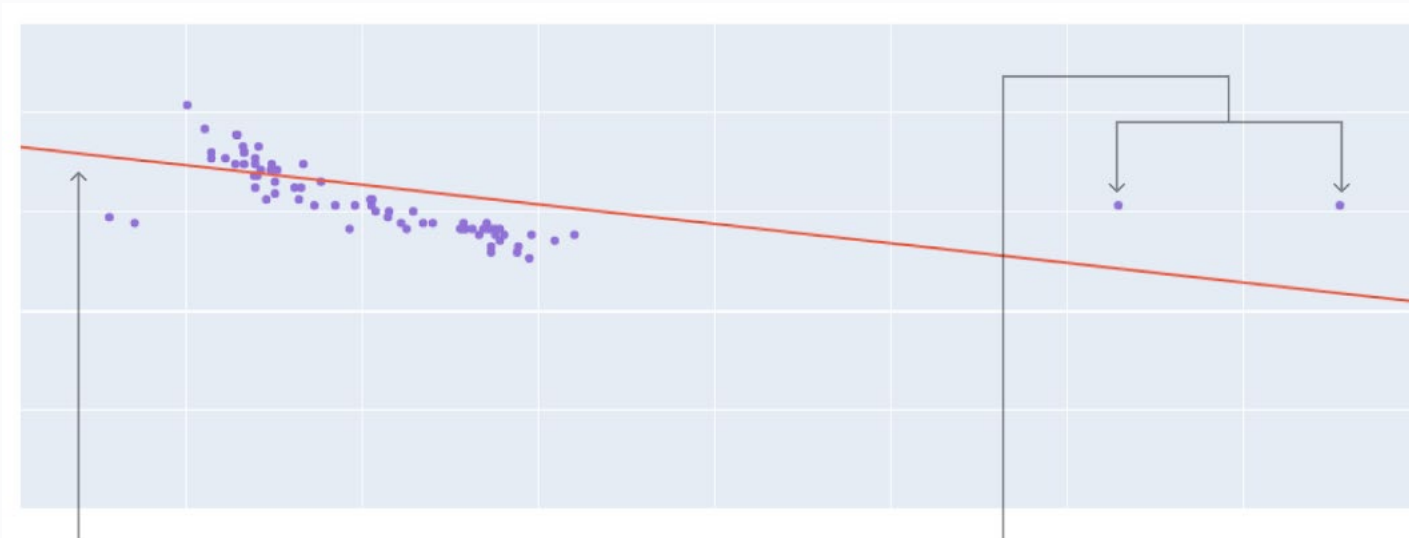
Linear regression: a first interpretable model

Loss is a numerical metric that describes how wrong a model's **predictions** are. Loss measures the distance between the model's predictions and the actual labels. The goal of training a model is to minimize the loss, reducing it to its lowest possible value.

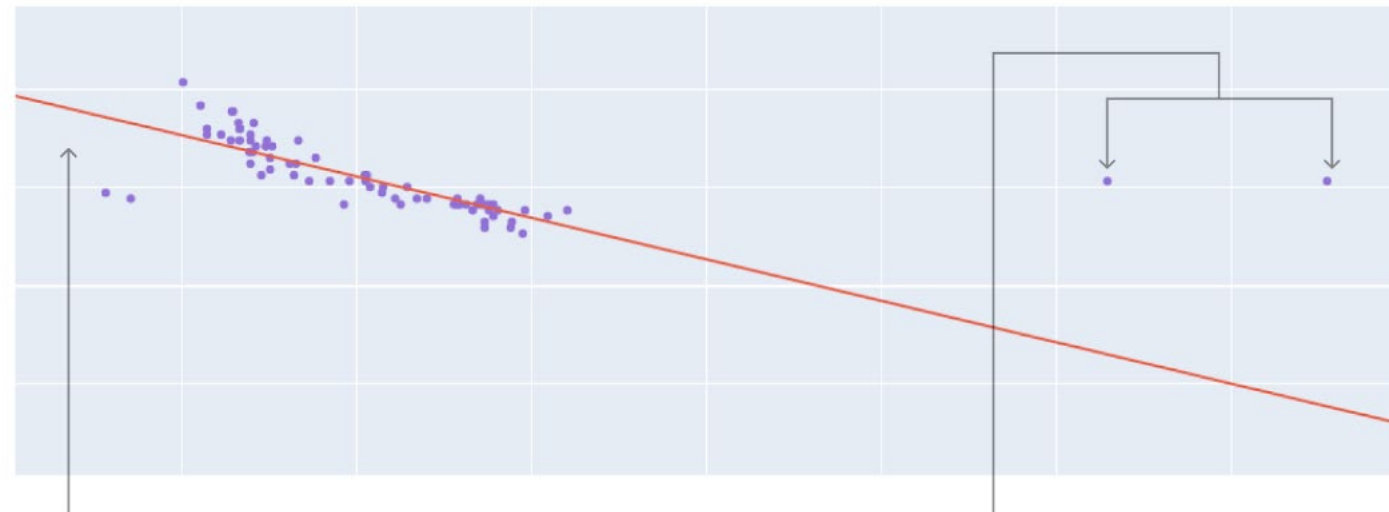


The arrows show how far the model's predictions are from the actual values.

Linear regression: a first interpretable model



Model trained
using MSE



Model trained
using MAE

Outliers

Linear regression: a first interpretable model

$$w_j := w_j + \text{error} \times x_j^{[i]}$$
$$b := b + \text{error}$$

 x_1

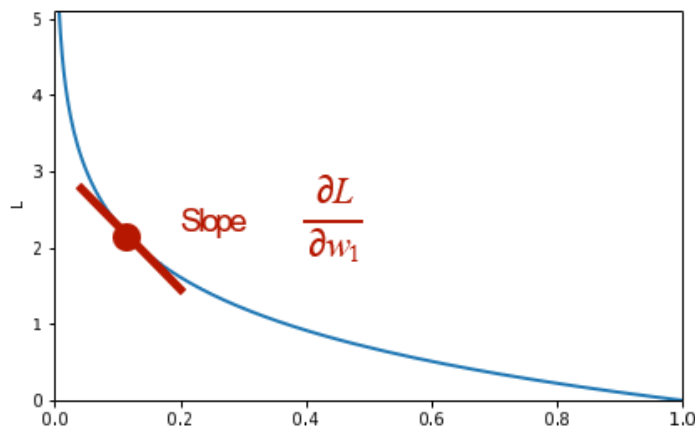
$w_1 := 0$

Compute
value

$b := 0$

Make Prediction

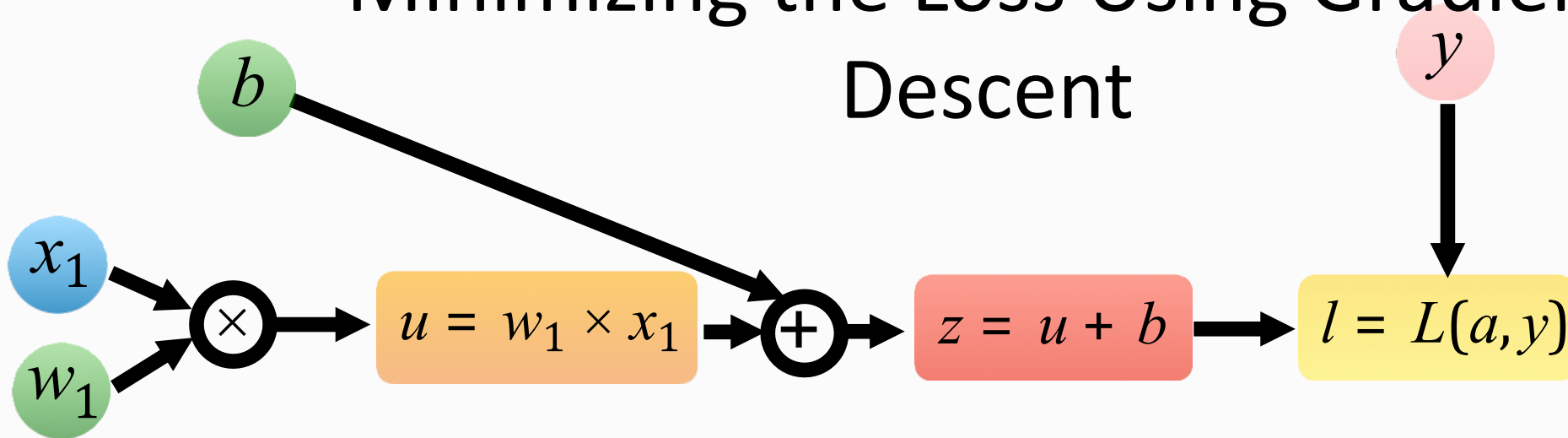
Update the weights based on slope of Loss Function



Pounds in 1000s (feature)	Miles per gallon (label)
3.5	18
3.69	15
3.44	18
3.43	16
4.34	15
4.42	14
2.37	24

Linear regression: a first interpretable model

Minimizing the Loss Using Gradient Descent



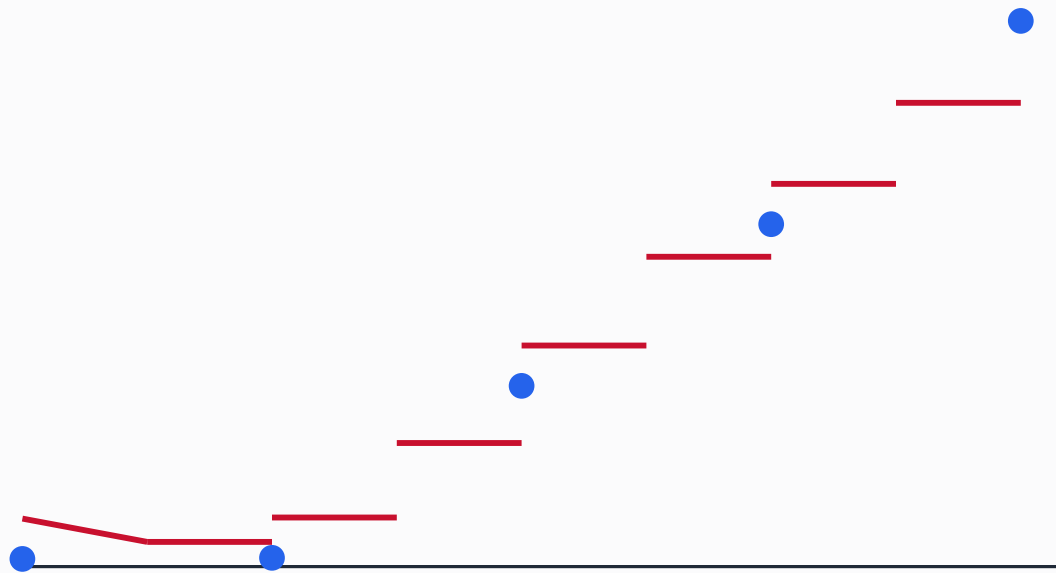
$$\frac{\partial L}{\partial w_1} = \frac{\partial u}{\partial w_1} \times \frac{\partial z}{\partial u} \times \frac{dL}{dz}$$

$$\frac{\partial L}{\partial b} = \frac{\partial z}{\partial b} \times \frac{dL}{dz}$$

$$\nabla L = \begin{bmatrix} \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial b} \end{bmatrix}$$

Polynomial terms: capturing simple nonlinearity

Polynomial regression still uses linear regression machinery, but adds transformed features such as x^2 .



Example model
 $\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$

When useful

Residuals show curved patterns, but the underlying relationship remains smooth and low-dimensional.

Risk

High-degree polynomials can chase noise and perform poorly on new data.

Regularization: controlling overly flexible models

Regularization adds a penalty for large coefficients to reduce overfitting and improve generalization.

Ridge regression

Shrinks coefficients toward zero.

Useful when many correlated features are present.

Lasso regression

Can shrink some coefficients exactly to zero.

Useful for feature selection and simpler models.

Tradeoff

Slightly more bias may reduce variance.

The goal is lower error on new data, not perfect fit to training data.

When would a smaller, more stable model be preferable to a slightly more accurate model?

Regression metrics: measure error in different ways

No single metric is universally best. Choose the metric based on the meaning and cost of prediction errors.

MAE**Average absolute error**

Easy to interpret in target units

RMSE**Square root of average squared error**

Penalizes large errors more strongly

 R^2

Fraction of variance explained relative to baseline

Useful summary but can hide large errors

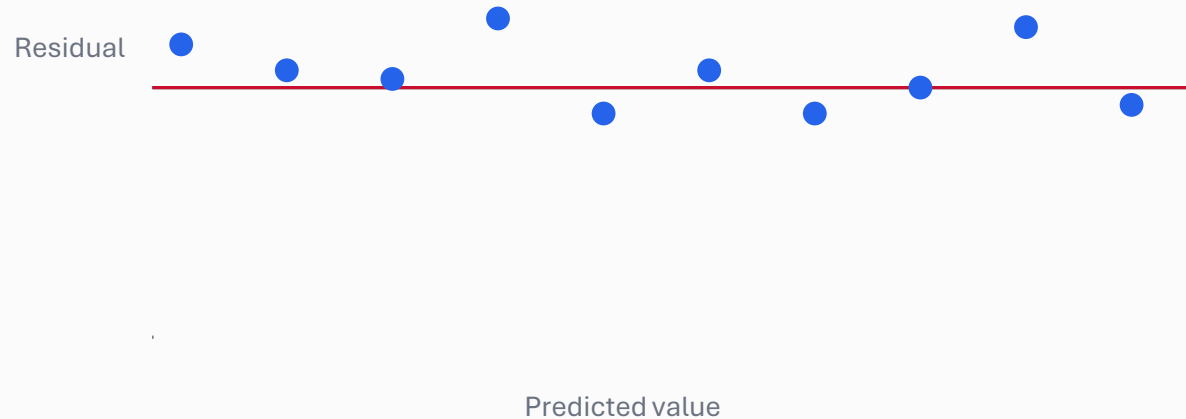
Domain threshold Percent of predictions within an acceptable error band

Often most meaningful to stakeholders

Metric choice is an ethical and practical design choice: it determines which errors the model prioritizes.

Residuals: where the model is wrong

A residual is observed value minus predicted value. Residual plots reveal patterns that a single metric can hide.



Look for patterns

- Centered around zero
- Curved pattern → missing nonlinearity
- Fan shape → changing variance
- Large isolated points → possible outliers
- Groups of errors → missing feature or population shift

Activity: compare candidate regression models

Use a small dataset to compare a baseline, a linear model, and a polynomial or regularized model.



Is the small gain from Model C worth the added complexity and lower interpretability?

Session 3 technical skill: minimal regression workflow

Goal: demonstrate split → baseline → fit → predict → evaluate for a continuous target.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

X = df[features]
y = df["strength_mpa"]
X_train, X_test, y_train, y_test = train_test_split(X, y, random_state=42)
```

```
baseline = np.full(len(y_test), y_train.mean())
model = LinearRegression()
model.fit(X_train, y_train)
pred = model.predict(X_test)

mae = mean_absolute_error(y_test, pred)
r2 = r2_score(y_test, pred)
```

- Baseline answers: “How well can we do with no model?”
- MAE answers: “How wrong are we in the original units?”
- R^2 answers: “How much variance is explained relative to the mean?”

Session 3 notebook

03_session3_regression_sklearn.ipynb

Regression interpretation: connect metrics to units

Keep the conversation grounded in the disciplinary meaning of the target variable.

Target	Material strength
Target unit	MPa
Metric	Mean Absolute Error
Interpretation	Typical absolute prediction error in MPa

Discussion questions

- Did the model outperform the baseline?
- Is the MAE practically acceptable?
- Are inputs available before prediction?
- What variable might be missing?

```
Baseline MAE: 18.4 MPa
Model MAE:    7.9 MPa
```

```
Question: Is 7.9 MPa useful for this application?
```

Discussion prompts

When is a linear model sufficient?

What does an error of 5 units mean in the discipline?

Should the model with the highest R^2 always be selected?

What residual pattern would make you distrust the model?

Can regression results be interpreted as causal evidence?

What baseline represents current practice?

Discussion goal: connect numerical performance to disciplinary usefulness and limits.

Session 3 synthesis and transition

Regression is useful when a faculty problem requires predicting a continuous outcome and evaluating the magnitude of error.

- Start simple** Use a mean, median, or disciplinary baseline.
- Fit carefully** Compare simple, polynomial, and regularized models.
- Evaluate meaningfully** Choose metrics that match the cost of error.
- Diagnose error** Residuals reveal patterns hidden by averages.
- Report limits** Prediction is not the same as explanation or causation.

Next: Classification fundamentals — predicting categories and interpreting confusion matrices.